

**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK**

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|                       |                                                                                                                                                                                                                                                           |                      |                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|
| <b>Course Code:</b>   | <b>CSA12</b>                                                                                                                                                                                                                                              | <b>Course Title:</b> | <b>Computer Architecture for Multicore Systems</b>    |
| <b>CO Assessed:</b>   | <b>CO4</b>                                                                                                                                                                                                                                                | <b>Assessment:</b>   | <b>ASSESSMENT TOOL1-INDUSTRY PROBLEM SOLVING TASK</b> |
| <b>Weightage:</b>     | <b>30%</b>                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | <b>25</b>                                             |
| <b>CO4 Statement:</b> | <i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i> |                      |                                                       |
| <b>Student Name:</b>  |                                                                                                                                                                                                                                                           | <b>Reg.No.:</b>      |                                                       |
| <b>Department:</b>    |                                                                                                                                                                                                                                                           | <b>Date:</b>         |                                                       |

**ANNEXURE A**

**INDUSTRY**

**PROBLEM SOLVING TASK**

- Smartphone Performance Optimization**  
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
- Cloud Server Memory Bottleneck**  
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
- Autonomous Vehicle Data Processing**  
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
- AI Inference Edge Device**  
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
- High-Performance Gaming Console**  
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

**Code of Conduct:**

*I \_\_Gayathri B.R(192521330)\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

## **Answers**

### **Q1. Smartphone Performance Optimization**

#### **Problem**

A smartphone manufacturer observes lag while switching between applications.

The main reason can be inefficient memory access and insufficient cache utilization. The system uses **L1, L2, L3 cache and DRAM**, each having different latency and capacity characteristics.

#### **Analysis**

- **L1 Cache:** Smallest and fastest cache. It stores frequently used instructions and data.
- **L2 Cache:** Larger than L1 but slightly slower. It acts as an intermediate cache.
- **L3 Cache:** Larger and slower than L1 and L2. It can be shared among processor cores.
- **DRAM:** Provides large-capacity main memory but has much higher access latency than cache.

When switching between applications, frequently accessed data should remain in cache. If the required data is not available in the cache, the processor must access DRAM, increasing latency.

#### **Proposed Memory Hierarchy**

1. Increase effective L2/L3 cache capacity.

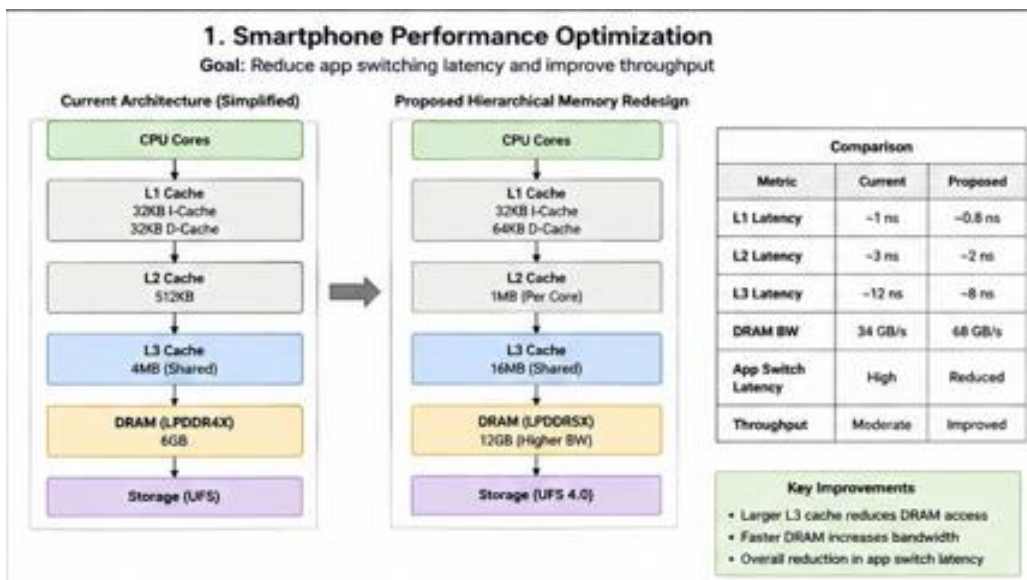
2. Improve cache hit rate using better replacement policies.
3. Keep frequently accessed application data in cache.
4. Use high-speed DRAM with sufficient bandwidth.
5. Optimize memory management to reduce unnecessary DRAM accesses.

## Expected Result

The redesigned hierarchy reduces average memory access latency and improves application switching performance. Frequently used data is served from faster cache levels, while DRAM is used mainly for larger data requirements.

## Conclusion

An efficient L1-L2-L3-DRAM hierarchy can reduce smartphone application-switching lag. Increasing cache effectiveness and reducing unnecessary DRAM accesses improves both latency and throughput.



## Q2. Cloud Server Memory Bottleneck

### Problem

A cloud service provider experiences high latency during database queries. The solution requires comparing **cache hierarchy** and **virtual memory paging** and selecting appropriate memory technologies.

### Analysis

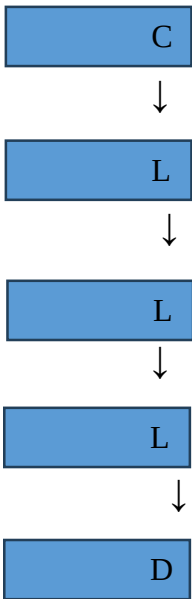
Database queries frequently access the same instructions and data. Cache memory can store frequently accessed database information close to the processor.

Virtual memory uses paging to provide a larger address space. However, when required data is not present in physical memory, a page fault occurs and data must be retrieved from secondary storage, causing high latency.

**Cache vs Virtual Memory**

| Feature            | Cache Hierarchy           | Virtual Memory                 |
|--------------------|---------------------------|--------------------------------|
| Main purpose       | Reduce memory access time | Extend available memory        |
| Speed              | Very high                 | Lower                          |
| Location           | Close to CPU              | Uses RAM and secondary storage |
| Latency            | Very low on cache hit     | High during page fault         |
| Management         | Hardware                  | Operating system               |
| Effect on database | Improves query response   | Provides larger memory space   |

**Recommended Solution**





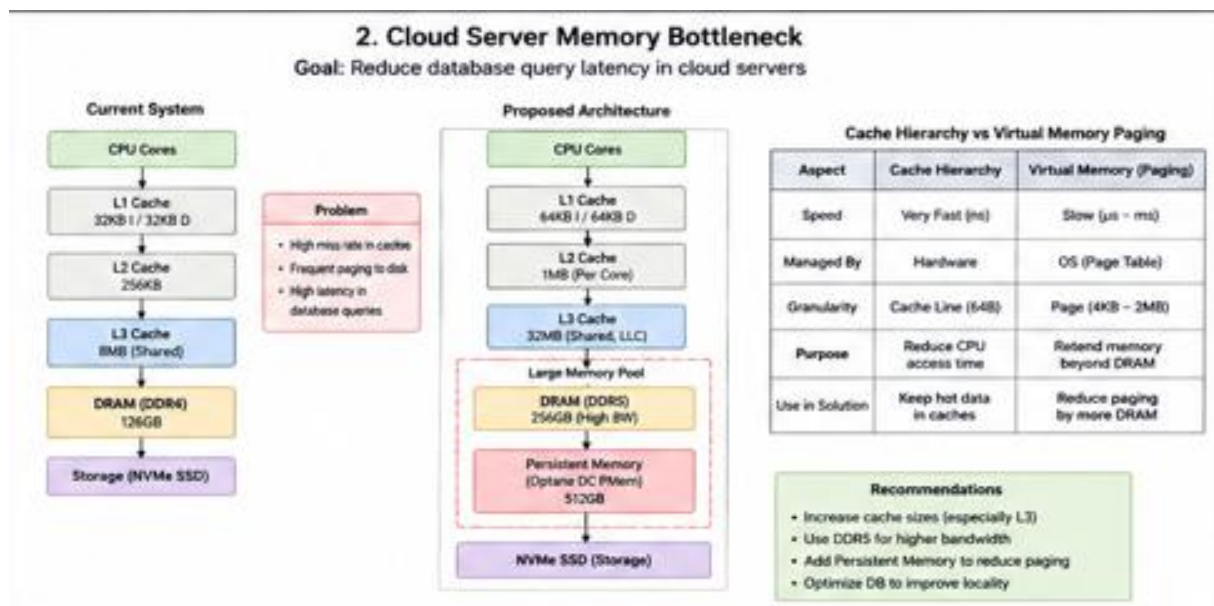
The server should use an effective cache hierarchy and sufficient DRAM to minimize page faults.

## Improvements

- Increase L3 cache effectiveness.
- Provide sufficient DRAM for active database workloads.
- Optimize database caching.
- Reduce unnecessary paging.
- Use high-speed SSD storage for unavoidable page operations.

## Conclusion

Cache hierarchy should be prioritized for reducing CPU access latency, while sufficient DRAM should be provided to minimize virtual-memory page faults. This combination can reduce database-query latency and improve cloud-server throughput.



## Q3. Autonomous Vehicle Data Processing

### Problem

An autonomous vehicle must process sensor data in real time. The memory system must provide **high bandwidth and low latency**. The required technologies are SRAM, DRAM and MRAM.

### Analysis

Autonomous vehicles continuously process data from cameras, radar, LiDAR and other sensors. Delays in processing can affect real-time decision making.

- **SRAM:** Very fast and suitable for frequently accessed sensor-processing data.
- **DRAM:** Provides larger capacity and is suitable for storing larger working datasets.
- **MRAM:** Non-volatile and provides fast access with good energy efficiency.

### Proposed Hierarchy



### Memory Allocation

| Memory | Suggested Use                                                |
|--------|--------------------------------------------------------------|
| SRAM   | Critical real-time data and frequently accessed instructions |
| DRAM   | Large sensor datasets and active applications                |
| MRAM   | Persistent/critical data requiring fast access               |

### Improvements

1. Store time-critical data in SRAM.
2. Use high-bandwidth DRAM for large sensor datasets.

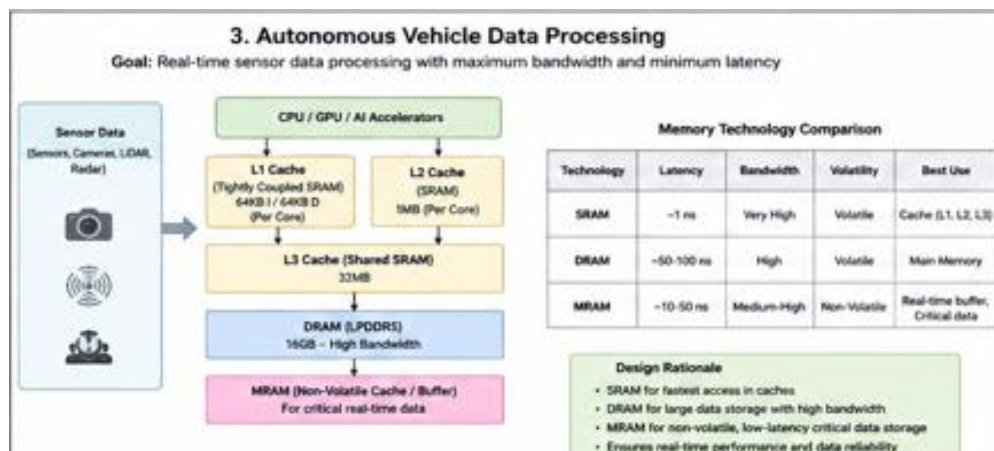
3. Use MRAM for selected non-volatile data.
4. Reduce unnecessary data movement between memory levels.
5. Use parallel memory access where possible.

### Expected Result

Frequently accessed and time-critical information can be accessed from SRAM with minimum latency, while DRAM provides the required capacity and bandwidth. MRAM provides persistent storage with fast access.

### Conclusion

A combined SRAM-DRAM-MRAM hierarchy provides a balance between **latency, bandwidth, capacity and energy efficiency**, making it suitable for real-time autonomous vehicle data processing.



## Q4. AI Inference Edge Device

### Problem

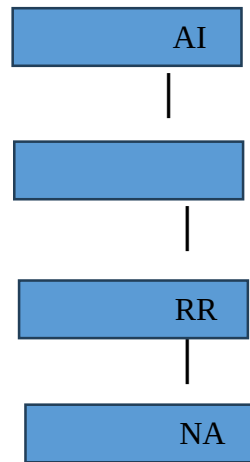
An edge AI device requires fast AI-model loading while maintaining low power consumption. The required technologies are NAND Flash, DRAM and RRAM.

### Analysis

AI models can be large and require fast access during inference.

- **NAND Flash:** Non-volatile and suitable for permanent storage of AI models.
- **DRAM:** Fast working memory for actively running AI models.
- **RRAM:** Non-volatile technology with potential for fast and energy-efficient memory operations.

## Proposed Hierarchy



## Memory Usage

**NAND Flash:** Stores the complete AI model permanently.

**RRAM:** Can store frequently used model parameters or selected data with low-power operation.

**DRAM:** Loads the active model and intermediate data during inference.

## Improvements

1. Store the complete model in NAND Flash.
2. Load frequently used model components into faster memory.
3. Use DRAM for active inference operations.
4. Use RRAM where non-volatile, energy-efficient storage is beneficial.
5. Reduce repeated loading from NAND Flash.

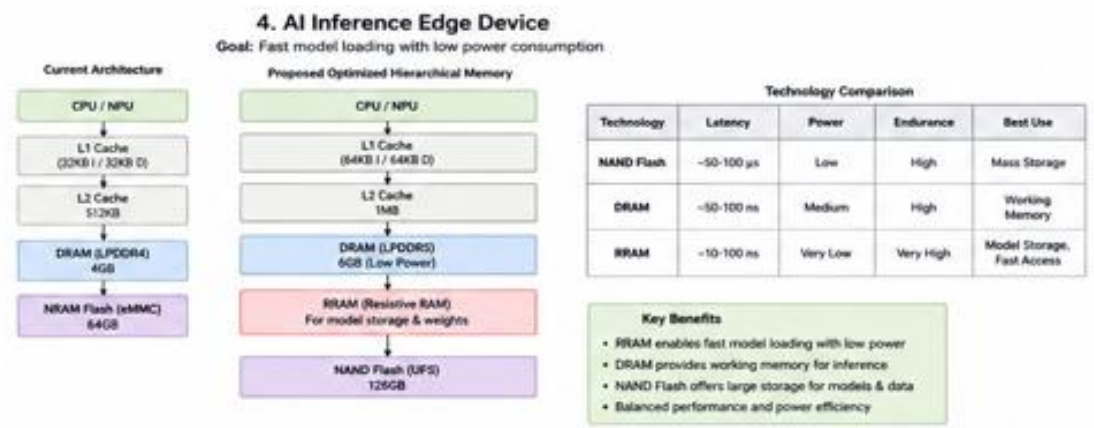
## Expected Result

The system reduces model-loading time by keeping frequently required information in faster memory and reduces energy consumption by minimizing unnecessary data transfers.

## Conclusion

A combination of NAND Flash, RRAM and DRAM provides a suitable hierarchy for edge AI devices by balancing **storage capacity, access speed and power efficiency**.





## Q5. High-Performance Gaming Console

### Problem

A gaming console experiences frame drops while loading large scenes. The problem requires analysis of **L3 cache and DRAM throughput** and improvements to the memory hierarchy.

### Analysis

Large game scenes contain textures, models, animations and other data. During scene loading, large amounts of data must be transferred between storage, DRAM, cache and the processor.

### L3 Cache

L3 cache is faster than DRAM and can provide frequently accessed data to processor cores with lower latency. However, its capacity is limited compared with DRAM.

### DRAM

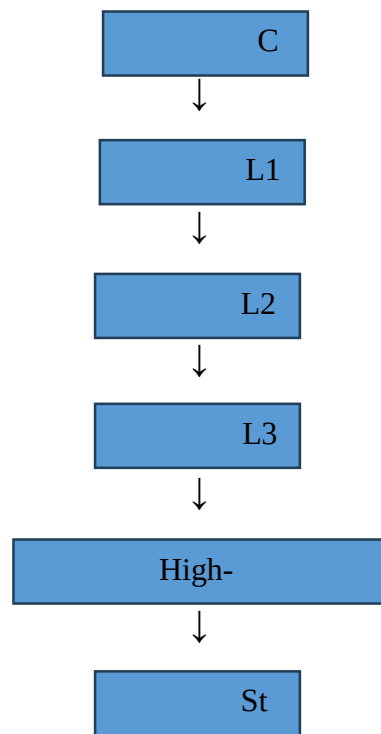
DRAM provides much larger capacity and is suitable for storing active game assets. However, it has higher latency than cache.

### Comparison

| Feature  | L3 Cache | DRAM   |
|----------|----------|--------|
| Latency  | Lower    | Higher |
| Capacity | Smaller  | Larger |

|                     |                          |                        |
|---------------------|--------------------------|------------------------|
| Speed               | Faster                   | Slower                 |
| Purpose             | Frequently accessed data | Large working datasets |
| Processor proximity | Very close               | Further from CPU       |

### Proposed Hierarchy



### Improvements

1. Improve L3 cache hit rate.
2. Use high-bandwidth DRAM.
3. Preload frequently required game assets.
4. Reduce unnecessary movement of data between memory levels.
5. Use efficient memory allocation for large textures and game objects.
6. Use parallel data transfers where possible.

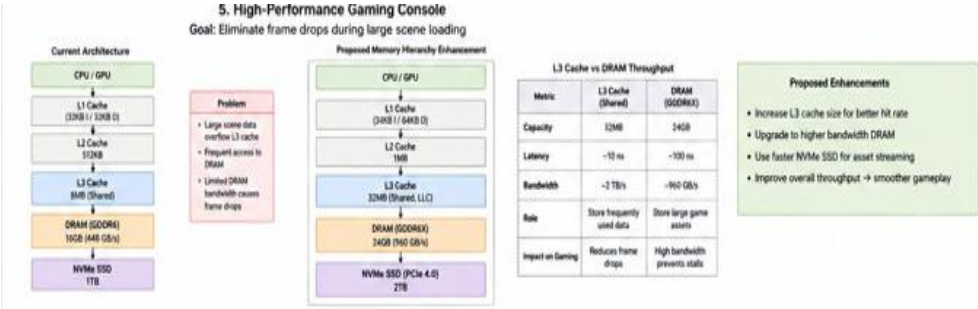
### Expected Result

Frequently accessed game data can be supplied quickly from cache, while high-bandwidth DRAM handles large scene data. This reduces memory bottlenecks and helps maintain stable frame rates.

Conclusion

Combining an efficient L3 cache with high-bandwidth DRAM can reduce scene-loading delays and minimize frame drops. A well-designed memory hierarchy improves both latency and throughput for gaming workloads.

These solutions directly address the CO4 requirement of designing hierarchical memory systems using **L1, L2, L3, DRAM, SRAM, NAND Flash, MRAM and RRAM** based on latency, bandwidth and throughput.



MARKS ALLOCATION

|    | Understanding of Industry Problem (20%) | Application of Engineering Concepts (25%) | Solution Design (25%) | Use of Tools (15%) | Analysis (10%) |
|----|-----------------------------------------|-------------------------------------------|-----------------------|--------------------|----------------|
| Q1 |                                         |                                           |                       |                    |                |
| Q2 |                                         |                                           |                       |                    |                |
| Q3 |                                         |                                           |                       |                    |                |
| Q4 |                                         |                                           |                       |                    |                |
| Q5 |                                         |                                           |                       |                    |                |

RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|----------|-----------|---------------------|----------------------|----------------------|---------------------|
|----------|-----------|---------------------|----------------------|----------------------|---------------------|

|                                     |     |                                                                              |                                                      |                                                |                                                   |
|-------------------------------------|-----|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20% | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15% | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10% | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |

